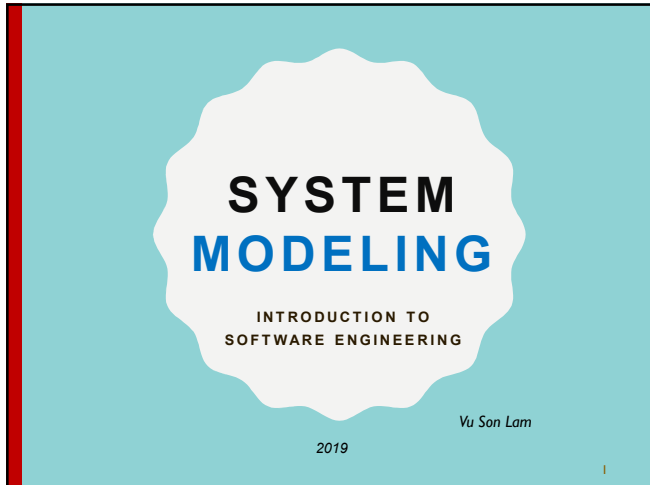




## Objectives

- The aim of this chapter is to introduce **some types of system model** that may be developed as part of the **requirements engineering** and **system design** processes.

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## Objectives

- When you have read the chapter, you will:
  - understand **how graphical models** can be used to **represent** software systems;
  - understand why **different types of model** are required and the fundamental system modeling perspectives of context, interaction, structure, and behavior;
  - have been introduced to some of the **diagram types in the UML** and how these diagrams may be used in system modeling;
  - be aware of the ideas underlying **model-driven engineering**, where a system is automatically generated from structural and behavioral models.

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## What is system modeling?

- **System modeling** is the **process of developing abstract models of a system**, with each model presenting a different view or perspective of that system.
- System modeling has now come to mean **representing a system** using some kind of graphical notation, which is now almost always based on notations in the Unified Modeling Language (UML).
- **System modelling helps**
  - the analyst to **understand** the **functionality** of the system and models are used to **communicate** with **customers**.

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## Existing and planned system models

- Models of the existing system are used during requirements engineering.
  - They help **clarify** what the existing system does
  - and can be **used as a basis** for discussing its strengths and weaknesses. These then lead to requirements for the new system.
- Models of the new system
  - are used during requirements engineering to **help explain** the **proposed requirements** to other system stakeholders.
  - Engineers use these models to **discuss design proposals** and to document the system for implementation.

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## Existing and planned system models

- In a **model-driven engineering process**, it is possible to **generate** a complete or partial system implementation from the system model.

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## System perspectives

- An **external perspective**, where you model the context or environment of the system.
- An **interaction perspective**, where you model the interactions between a system and its environment, or between the components of a system.
- A **structural perspective**, where you model the organization of a system or the structure of the data that is processed by the system.
- A **behavioral perspective**, where you model the dynamic behavior of the system and how it responds to events.

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## Use of graphical models

- There are three ways in which graphical models are commonly used:
  - As a means of **facilitating discussion** about an existing or proposed system.
  - As a way of **documenting an existing system**.
  - As a **detailed system description** that can be used to generate a **system implementation**.

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## Use of graphical models

- As a means of facilitating discussion about an existing or proposed system
  - **Incomplete** and **incorrect** models are OK as their role is to support discussion.
- As a way of documenting an existing system
  - Models should be an **accurate** representation of the system but **need not be complete**.
- As a detailed system description that can be used to generate a system implementation
  - Models have to be both **correct** and **complete**.

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## UML overview

- UML (**Unified Modeling Language**) là ngôn ngữ mô hình hóa hợp nhất, dùng để trực quan hóa thiết kế của hệ thống.
- UML được xây dựng dựa trên sự hợp nhất của các phương pháp **Booch**, **OMT** (object-modeling technique, Rumbaugh), **OOSE** (object-oriented software engineering, Jacobsen) vào những năm 1994-1995.
- Năm 1997, UML được chấp nhận như một chuẩn chính thức bởi tổ chức **OMG** (Object Management Group) → **UML 1.1**

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## UML overview

- Năm 2005: **UML 2.0** được công bố.
- Năm 2005, UML được công bố như một chuẩn ISO (International Organization for Standardization)
- Mới nhất, 12/2017, **UML 2.5.1** được công bố.
- Hiện UML vẫn được quản lý bởi **OMG**:  
<https://www.omg.org/spec/UML/About-UML/>



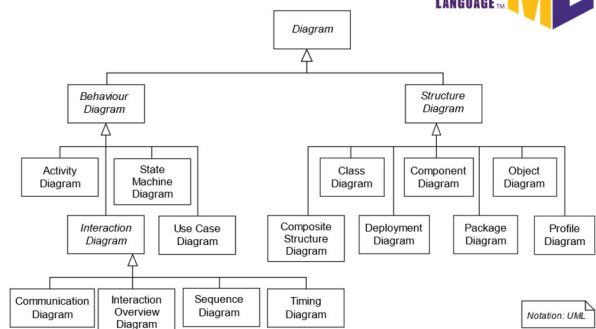
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## Some UML diagram types

- Activity diagrams**, which show the activities involved in a process or in data processing.
- Use case diagrams**, which show the interactions between a system and its environment.
- Sequence diagrams**, which show interactions between actors and the system and between system components.
- Class diagrams**, which show the object classes in the system and the associations between these classes.
- State diagrams**, which show how the system reacts to internal and external events.

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## UML diagram types



Source: [https://en.wikipedia.org/wiki/Unified\\_Modeling\\_Language](https://en.wikipedia.org/wiki/Unified_Modeling_Language)

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## Context models

- Context models are used to illustrate the operational context of a system - they show what lies outside the system boundaries.
- At an early stage in the specification of a system, you should decide on the **system boundaries**.
- This involves working with system stakeholders to decide
  - what functionality should be included in the system and
  - what is provided by the system's environment.

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## Context models

- System boundaries:
  - You may decide that automated support for some business processes should be implemented but others should be manual processes or supported by different systems.
  - You should look at possible overlaps in functionality with existing systems and decide where new functionality should be implemented.

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## Context models

- For example, say you are developing the specification for the patient information system for mental healthcare.
- This system is intended to manage information about patients attending mental health clinics and the treatments that have been prescribed.
- In developing the specification for this system, you have to decide whether the system should focus exclusively on collecting information about consultations (using other systems to collect personal information about patients) or whether it should also collect personal patient information.

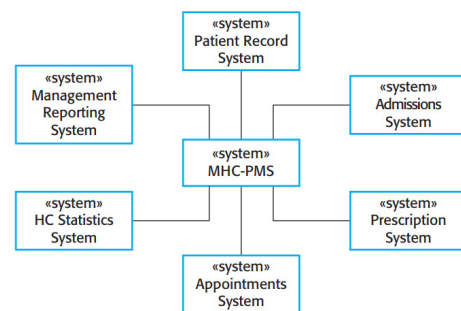
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## Context models

- The advantage of relying on other systems for patient information is that you avoid duplicating data.
- The major disadvantage, however, is that using other systems may make it slower to access information. If these systems are unavailable, then the MHC-PMS cannot be used.

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## Context models



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## Context models

- Context models normally show that the environment includes several other automated systems. However, they do not show the types of relationships between the systems in the environment and the system that is being specified.
  - External systems might produce data for or consume data from the system.
  - They might share data with the system, or they might be connected directly, through a network or not connected at all.
  - They might be physically co-located or located in separate buildings.
- All of these relations may affect the requirements and design of the system being defined and must be taken into account.

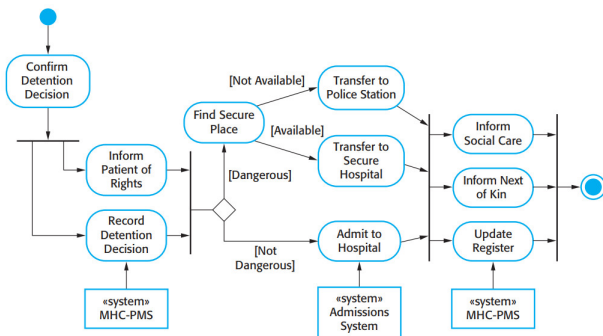
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## Context models

- Therefore, simple context models are used along with other models, such as business process models. These describe human and automated processes in which particular software systems are used.

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## Context models



UML activity diagram: Process model of involuntary detention

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## Interaction models

- All systems involve interaction of some kind:
  - User interaction, which involves user inputs and outputs
  - Interaction between the system being developed and other systems
  - Interaction between the components of the system.

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## Interaction models

- Modeling user interaction is important as it helps to identify user requirements.
- Modeling system to system interaction highlights the communication problems that may arise.
- Modeling component interaction helps us understand if a proposed system structure is likely to deliver the required system performance and dependability.

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## Interaction models

- Some approaches to interaction modeling:
  - **Use case modeling**, which is mostly used to model interactions between a system and external actors (users or other systems).
  - **Sequence diagrams**, which are used to model interactions between system components, although external agents may also be included.

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## Interaction models

- Use case models and sequence diagrams present interaction at different levels of detail and so may be used together.
  - The details of the interactions involved in a high-level use case may be documented in a sequence diagram.
- The UML also includes communication diagrams that can be used to model interactions.
  - They are an alternative representation of sequence charts.
  - In fact, some tools can generate a communication diagram from a sequence diagram.

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## Use case modeling

- Use case modeling was originally developed by Jacobson et al. (1993) in the 1990s and was incorporated into the first release of the UML.
- Use case modeling is widely used to support **requirements elicitation**. A use case can be taken as a **simple scenario** that describes **what a user expects** from a system.
- Each use case represents a **discrete task** that involves **external interaction** with a system.

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## Use case modeling



Transferdata use case

Formally, use case diagrams should use lines without arrows.

However, the arrows are used informally to indicate that the medical receptionist initiates the transaction and data is transferred to the patient record system

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## Use case modeling

- Use case diagrams give a fairly simple overview of an interaction so you have to provide more detail to understand what is involved.
- This detail can either be
  - a simple textual description
  - a structured description in a table,
  - a sequence diagram.
  - an activity diagram.
- You chose the most appropriate format depending on the use case and the level of detail that you think is required in the model.

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## Use case modeling

MHC-PMS: Transfer data

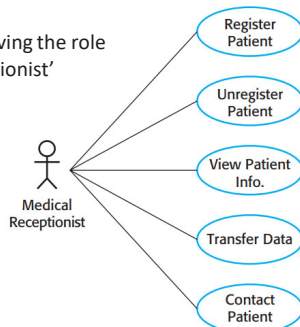
|             |                                                                                                                                                                                                                                                                                              |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Actors      | Medical receptionist, patient records system (PRS)                                                                                                                                                                                                                                           |
| Description | A receptionist may transfer data from the MHC-PMS to a general patient record database that is maintained by a health authority. The information transferred may either be updated personal information (address, phone number, etc.) or a summary of the patient's diagnosis and treatment. |
| Data        | Patient's personal information, treatment summary                                                                                                                                                                                                                                            |
| Stimulus    | User command issued by medical receptionist                                                                                                                                                                                                                                                  |
| Response    | Confirmation that PRS has been updated                                                                                                                                                                                                                                                       |
| Comments    | The receptionist must have appropriate security permissions to access the patient information and the PRS.                                                                                                                                                                                   |

Tabular description of the 'Transfer data' use case

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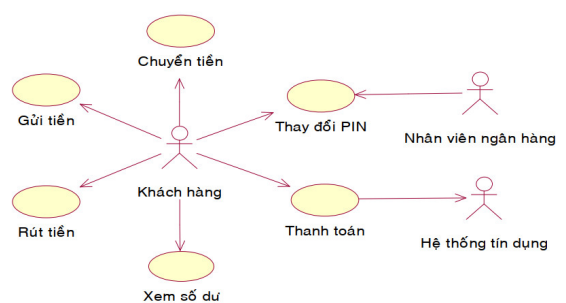
## Use case modeling

Use cases involving the role 'medical receptionist'



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## Use case modeling



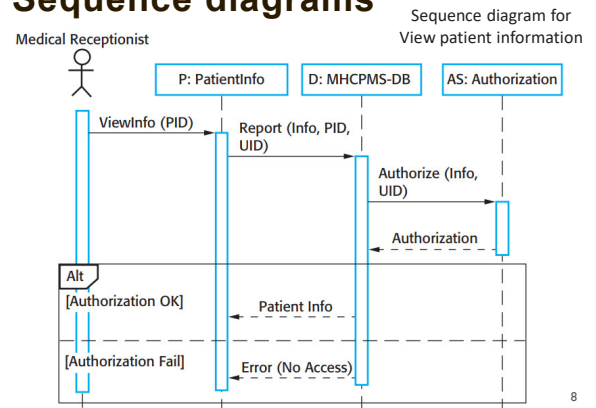
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## Sequence diagrams

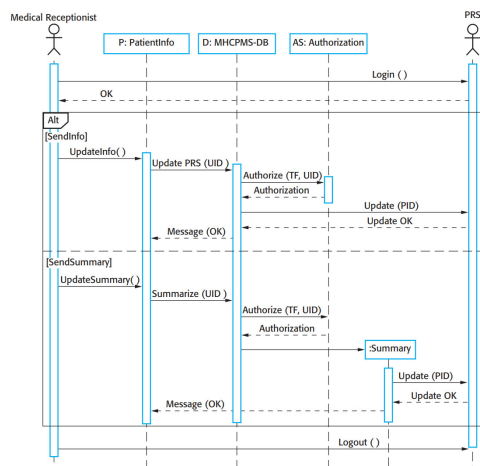
- Sequence diagrams in the UML are primarily used to model the interactions between the **actors** and the **objects** in a system and the interactions between the **objects themselves**.
- A sequence diagram shows the sequence of interactions that take place during a particular use case or use case instance.

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## Sequence diagrams



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## Sequence diagrams

- Unless you are using sequence diagrams for code generation or detailed documentation, you don't have to include every interaction in these diagrams.
- If you develop system models early in the development process to support requirements engineering and high-level design, there will be many interactions which depend on implementation decisions.
  - For example, in the example above, the decision on how to get the **user's identifier** to check authorization is one that **can be delayed**.
  - In an implementation, this might involve **interacting with a User object** but this is not important at this stage and so **need not be included** in the sequence diagram.

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## Structural models

- Structural models of software display the **organization** of a system in terms of the **components** that make up that system and **their relationships**.
- Structural models may be
  - static models, which show the structure of the system design. (e.g **class diagrams**)
  - or dynamic models, which show the organization of the system when it is executing.

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## Structural models

- You create structural models of a system when you are **discussing** and **designing** the **system architecture**.
- Architectural design** is a particularly important topic in software engineering and UML **component**, **package**, and **deployment** diagrams may all be used when presenting architectural models.

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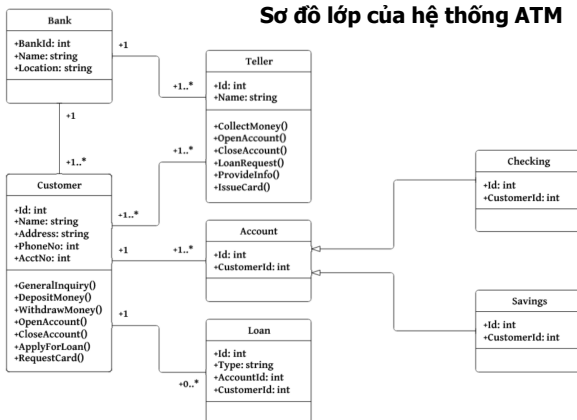
## Class diagrams



- Class diagrams are used when developing an object-oriented system model to show the classes in a system and the associations between these classes.
- Class diagrams in the UML can be expressed at different levels of detail.
  - When you are developing a model, the first stage is usually to look at the world, identify the essential objects, and represent these as classes.

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### Sơ đồ lớp của hệ thống ATM



[https://medium.com/@smagid\\_allThings/uml-class-diagrams-tutorial-step-by-step-520fd83b300b](https://medium.com/@smagid_allThings/uml-class-diagrams-tutorial-step-by-step-520fd83b300b)

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## Behavioral models

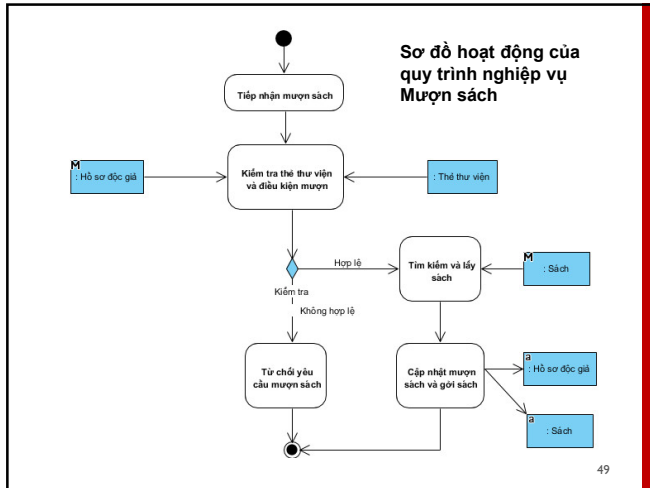
- Behavioral models are models of the dynamic behavior of the system as it is executing.
- They show what happens or what is supposed to happen when a system responds to a stimulus from its environment.
- You can think of these stimuli as being of two types:
  - Data** - Some data arrives that has to be processed by the system.
  - Events** - Some event happens that triggers system processing. Events may have associated data but this is not always the case.

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## Data-driven modeling

- Data-driven models show the sequence of actions involved in processing input data and generating an associated output.
  - DFD (**data-flow diagrams**)
  - The UML does not support data-flow diagrams. UML 2.0 introduced **activity diagrams**, which are similar to data-flow diagrams.

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## Activity diagram

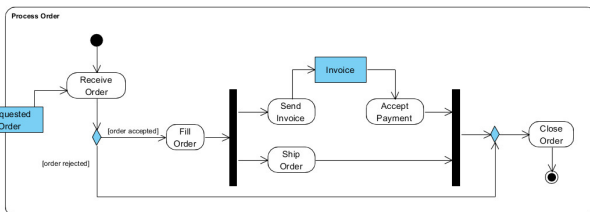
### • Bài tập:

Vẽ sơ đồ hoạt động cho quy trình **Xử lý đơn hàng**. Quy trình như sau:

- Khi có yêu cầu đặt hàng, nhân viên tiếp nhận đơn đặt hàng.
- Nhân viên kiểm tra xem đơn hàng có phù hợp không. Nếu không phù hợp → đóng đơn hàng → kết thúc quy trình.
- Nếu phù hợp, nhân viên lập đơn hàng.
- Sau đó, thực hiện song song gửi hàng và gửi hóa đơn.
- Khách hàng nhận hàng và thanh toán xong → đóng đơn hàng → kết thúc quy trình.

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## Activity diagram



## Event-driven modeling

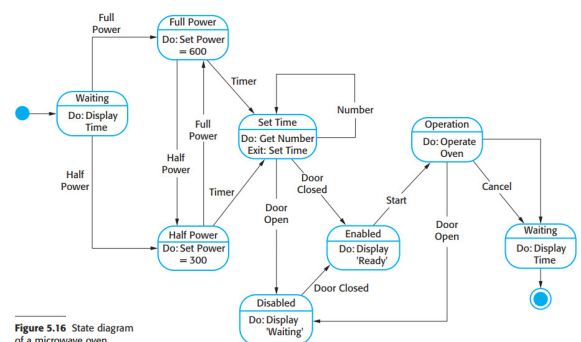
- Event-driven modeling shows how a system responds to external and internal events.
- It is based on the assumption that a system has a finite number of states and that events (stimuli) may cause a transition from one state to another.
  - For example, a system controlling a valve may move from a state 'Valve open' to a state 'Valve closed' when an operator command (the stimulus) is received. This view of a system is particularly appropriate for real-time systems.
- The UML supports event-based modeling using **state diagrams**.

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## State diagram



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| State      | Description                                                                                                                                                                                                                        |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waiting    | The oven is waiting for input.<br>The display shows the current time.                                                                                                                                                              |
| Half power | The oven power is set to 300 watts.<br>The display shows 'Half power'.                                                                                                                                                             |
| Full power | The oven power is set to 600 watts.<br>The display shows 'Full power'.                                                                                                                                                             |
| Set time   | The cooking time is set to the user's input value.<br>The display shows the cooking time selected and is updated as the time is set.                                                                                               |
| Disabled   | Oven operation is disabled for safety. Interior oven light is on.<br>Display shows 'Not ready'.                                                                                                                                    |
| Enabled    | Oven operation is enabled. Interior oven light is off.<br>Display shows 'Ready to cook'.                                                                                                                                           |
| Operation  | Oven in operation. Interior oven light is on. Display shows the timer countdown. On completion of cooking, the buzzer is sounded for five seconds. Oven light is on.<br>Display shows 'Cooking complete' while buzzer is sounding. |

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| Stimulus    | Description                                    |
|-------------|------------------------------------------------|
| Half power  | The user has pressed the half-power button.    |
| Full power  | The user has pressed the full-power button.    |
| Timer       | The user has pressed one of the timer buttons. |
| Number      | The user has pressed a numeric key.            |
| Door open   | The oven door switch is not closed.            |
| Door closed | The oven door switch is closed.                |
| Start       | The user has pressed the Start button.         |
| Cancel      | The user has pressed the Cancel button.        |

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